



Review

Artificial reefs and reef restoration in the Laurentian Great Lakes

Mathew McLean¹, Edward F. Roseman^{*}, Jeremy J. Pritt, Gregory Kennedy, Bruce A. Manny

USGS Great Lakes Science Center, 1451 Green Road, Ann Arbor, MI 48105, USA

ARTICLE INFO

Article history:

Received 27 May 2014

Accepted 30 October 2014

Available online 19 December 2014

Communicated by Tom Stewart

Index words:

Artificial reefs

Habitat

Restoration

Monitoring

Spawning

ABSTRACT

We reviewed the published literature to provide an inventory of Laurentian Great Lakes artificial reef projects and their purposes. We also sought to characterize physical and biological monitoring for artificial reef projects in the Great Lakes and determine the success of artificial reefs in meeting project objectives. We found records of 6 artificial reefs in Lake Erie, 8 in Lake Michigan, 3 in Lakes Huron and Ontario, and 2 in Lake Superior. We found 9 reefs in Great Lakes connecting channels and 6 reefs in Great Lakes tributaries. Objectives of artificial reef creation have included reducing impacts of currents and waves, providing safe harbors, improving sport-fishing opportunities, and enhancing/restoring fish spawning habitats. Most reefs in the lakes themselves were incidental (not created purposely for fish habitat) or built to improve local sport fishing, whereas reefs in tributaries and connecting channels were more frequently built to benefit fish spawning. Levels of assessment of reef performance varied; but long-term monitoring was uncommon as was assessment of physical attributes. Artificial reefs were often successful at attracting recreational species and spawning fish; however, population-level benefits of artificial reefs are unclear. Stressors such as sedimentation and bio-fouling can limit the effectiveness of artificial reefs as spawning enhancement tools. Our investigation underscores the need to develop standard protocols for monitoring the biological and physical attributes of artificial structures. Further, long-term monitoring is needed to assess the benefits of artificial reefs to fish populations and inform future artificial reef projects.

Published by Elsevier B.V. on behalf of International Association for Great Lakes Research.

Contents

Introduction	1
Review of artificial reef projects in the Great Lakes	2
Purpose and design of artificial reefs in the Great Lakes	2
Biological monitoring of reef projects	4
Physical monitoring of artificial reef projects	5
Success of artificial reefs in the Great Lakes in attracting fish and increasing fish spawning	5
Stressors and degradation of artificial reefs	6
Limitations of artificial reef monitoring projects	6
Conclusions	6
Acknowledgments	6
References	6

Introduction

Habitat degradation is a global source of imperilment of fishes (e.g., Jelks et al., 2008; Munday, 2004; Powels et al., 2000), and restoration of physical habitat is often used to mitigate losses and degradation of

natural habitat (Bassett, 1994; Palmer et al., 2005). In large aquatic systems such as the Laurentian Great Lakes, artificial reefs are often created to provide spawning and nursery habitat to benthic-spawning fishes (e.g., Fitzsimons, 1996; Gannon, 1990; NOAA, 2007). Artificial reefs have been used as a fishery management tool for over 40 years in the Laurentian Great Lakes and have often been championed as a successful means for improving recreational fisheries (Kelch, 2012). However, unequivocal evidence of improved fisheries resulting from habitat restoration projects is often lacking (Hughes et al., 2014; Jähnig et al., 2011; Whiteway et al., 2010). We conducted a literature review of Great Lakes reef projects to

^{*} Corresponding author.

E-mail addresses: mattmclean12@gmail.com (M. McLean), eroseman@usgs.gov (E.F. Roseman).

¹ Current address: 621 North 10th Street, Plainwell, MI 49080.

guide future habitat restoration and monitoring in the Great Lakes and other large aquatic systems.

Artificial reefs have been constructed in freshwater and marine systems (1) to attract fish (Creque et al., 2006; Kelch, 2012) and (2) to provide spawning, nursery, and adult habitats, and increase fish abundance (Brickhill et al., 2005; Dumont et al., 2011; Kuhl, 1992; Pickering and Whitmarsh, 1997). Artificial reefs have been proven to attract fish and increase catch rates in recreational fisheries (Bombace et al., 1994; Brickhill et al., 2005; Kelch, 2012), but the ability of reefs to increase fish abundance is not well documented in freshwater and marine systems (Baine, 2001; Pickering and Whitmarsh, 1997). Several reports reviewed artificial reef projects in marine and freshwater systems and determined that reefs seldom increased fish population abundance. Problems arose from flawed study design (Brickhill et al., 2005), lack of clear objectives for reef building (Baine, 2001; Claudet and Pelletier, 2004), and inadequate monitoring of reefs (Baine, 2001). Few studies concluded that artificial reefs can perform as well or better than natural reefs in terms of increasing fish abundance and richness at the reef site (Carr and Hixon, 1997; Koeck et al., 2014). From an ecological perspective, whether artificial reefs can function as well as natural reefs remains unclear (Perkol-Finkel et al., 2006). Some artificial reefs have produced fish communities that are different in composition from those in natural habitats (Ambrose and Swarbrick, 1989; Flopp et al., 2013).

Despite careful planning, artificial reefs can be damaged or degrade over time. Damage to reefs can be caused by navigation and channelization, sedimentation, intrusion of groundwater, and changes in water levels (Rutherford et al., 2004). Other potential threats to reef quality and function over time include redistribution of reef materials by wave action, fouling by invasive species (e.g., Dreissenidae), and algae accumulation (Marsden and Chotkowski, 2001). These factors often limit the effectiveness of both artificial and natural reefs by filling interstices in the reef that protect fish eggs until they hatch. Most of these stressors do not pose an immediate threat, but they can develop over time. In order to quantify the contribution of man-made reefs to fish populations and provide a basis for comparison with natural reefs, routine standardized long-term monitoring protocols should be implemented (Gannon, 1990).

Here we review available information about artificial reef projects in the Laurentian Great Lakes and synthesize information on reef construction, monitoring, and performance. Our objectives were (1) to inventory artificial reef projects in the Great Lakes and their connecting channels and tributaries, (2) to characterize the purpose and construction methods of artificial reef projects, (3) to determine the type and extent of physical and/or biological monitoring on Great Lakes artificial reefs, and (4) to determine whether artificial reefs have achieved their intended objectives.

Review of artificial reef projects in the Great Lakes

We conducted a literature search using Google (<https://www.google.com/>), Google Scholar (<http://scholar.google.com/>), the U.S. Geological Survey Library (<http://rl3mq7xr4s.search.serialssolutions.com/>), ISI Web of Knowledge (http://go.isiprproducts.com/cgi-bin/MPL/2_0_1/User/Logon.cgi), and the online search function of the Journal of Great Lakes Research (<http://www.sciencedirect.com/>). We paired location keywords, including “Great Lakes,” “Lake Michigan,” “Lake Erie,” “Detroit River,” etc., with subject keywords such as “artificial reef,” “constructed reef,” and “spawning habitat restoration”. These terms were also searched without a specific location to obtain results for artificial reefs located outside of the Great Lakes. Lists of references in publications were searched forward (papers that cited the current paper) and backward (papers cited within) to find other relevant publications, including gray literature such as agency reports, theses, and news releases. For all relevant Great Lakes reef studies discovered, we obtained reef site name, location, year of construction, objectives for construction, materials used (Table 1), construction design, monitoring

scope, and timing/duration of monitoring. We also determined whether the objectives of each project were achieved, based on published accounts and personal communications with agency scientists familiar with respective projects.

Purpose and design of artificial reefs in the Great Lakes

In our literature search, we found 38 artificial reefs constructed in the Great Lakes basin between the late 1800s and 2013. We found at least one project in each of the Great Lakes, as well as several in the Great Lakes connecting channels (St. Clair-Detroit rivers and St. Lawrence River) and tributaries (Fig. 1). Some projects consisted of multiple reefs, and unless individual reefs within a project were differentiated from one another by name, a project or site referred to all reefs constructed at that location. Lake Michigan (8 projects) and the connecting channels (9 projects) had the greatest number of reef projects, whereas Lake Superior had the fewest reef projects (2 projects; Fig. 2). About 62% of man-made reefs have been the focus of biological monitoring but not physical monitoring, 24% of the reefs had both physical and biological monitoring, and about 14% had not been monitored (Fig. 2). Size and design of artificial reefs also varied on a project specific basis. Sizes of constructed structures ranged from less than 95 m² (Marsden et al., 1995) to over 33,000 m² (Yauck, 2009). Physical designs also varied from a 3-m high wall of rock in East Tawas Bay, Lake Huron (Foster and Kennedy, 1995), to individual uniform beds of various materials to assess preference for rock type in the Detroit River (Roseman et al., 2011) and large piles of rock in Lake Ontario (Marsden et al., 1995).

The earliest artificial reefs were created incidentally, for purposes other than fish habitat (Fig. 3). For example, the coal cinder reef in the North Channel of the St. Clair River is thought to have been formed in the late 1800s when ships discarded coal cinders while moored at the site (Nichols et al., 2003). Other examples of incidental reefs include projects aimed at abating physical impacts of currents and waves, providing safe harbors, or to protect water intake pipes for industry and municipalities (i.e., Liston et al., 1985; Marsden and Chotkowski, 2001; Peck, 1986; Rutecki et al., 1985). Interest in building reefs to improve sport fishing and restore functional fish spawning habitat started after 1980 (Fig. 3). Structures designed to improve recreational fishing opportunities were most often placed near metropolitan areas where anglers could have access (Bader, 1985; Kelch, 2012; Kelch et al., 1999; Kevern et al., 1985; Stewart and Haynes, 1994; Vincent, 1995). For example, large reefs were built to attract fishes such as walleye and smallmouth bass near the cities of Cleveland, Ohio (Kelch, 2012) and Chicago, Illinois (Creque et al., 2006). These projects had social and economic objectives (i.e., increase angler effort and success) rather than to increase fish populations. Sport-fishing reefs have been constructed from natural materials, such as limestone and granite (Creque et al., 2006), and man-made materials, including cement, brick, and even old stadium rubble (Kelch et al., 1999) (Table 1). Hence, reefs aimed at improving recreational fisheries have often been sited to provide easy access to anglers and were constructed out of convenient, available materials, which may have limited the value of these reefs as fish spawning habitat.

More recently, the focus of artificial spawning habitat construction shifted to restoring and increasing functional spawning habitat in order to enhance fish populations (e.g., Dumont et al., 2011; Fitzsimons, 1996; Manny et al., 2010; Roseman et al., 2011) (Fig. 3). Emphasis of these projects was on spawning habitat for imperiled or important species, such as lake sturgeon (e.g., Dumont et al., 2011; Roseman et al., 2011), lake trout (Fitzsimons, 1996), and walleye (Geiling et al., 1996), or improving diversity of benthic-spawning fishes (Manny et al., 2010). Material used to construct fish spawning reefs has most frequently been broken limestone (Roseman et al., 2011) but has also included igneous rocks (Geiling et al., 1996) (Table 1). Substrate size also varied across artificial reef projects, ranging from boulders (Marsden et al., 1995) to cobble and gravel (Dumont et al., 2011). Recent projects used a large

Table 1

Artificial reefs located in the Laurentian Great Lakes, tributaries the St. Clair and Detroit River System (SCDRS), and St. Lawrence River (SL) connecting channels.

Type	Reef site	Year created	Material	Purpose	Biological assessment	Physical assessment	Reference
Connecting channels	Belle Isle (SCDRS)	2004	Limestone, cobble, coal cinders	Fish Spawning	4 years	None	Manny (2006)
	Fighting Island (SCDRS)	2008	Limestone, fieldstone	Fish spawning	Ongoing	None	Roseman et al. (2011)
	Northeast Belle Isle (SCDRS)	2013	Limestone, fieldstone	Fish spawning	None yet	None yet	G. Kennedy (personal communication)
	Algonac (SCDRS)	Late 1800s	Coal cinders	Incidental reef	2 years but ongoing	2 years	Nichols et al. (2003); E. Roseman (personal communication)
	Middle Channel (SCDRS)	2012	Limestone, fieldstone	Fish spawning	Ongoing	None	E. Roseman (personal communication)
	Port Huron (SCDRS)	2012	Limestone	Fish spawning	None	None	St. Clair River Bi-National Public Advisory Council (2013)
	Sythe Energies (SL)	1993	Limestone	Incidental reef	3 years	Diver visual	Johnson et al. (2006)
	Iroquois S. (SL)	2007	Gravel and boulders	Fish spawning	2 years	1 year	Environment Illimité Inc. (2009, 2010)
	Iroquois N. (SL)	2007	Gravel and boulders	Fish spawning	2 years	1 year	Environment Illimité Inc. (2009, 2010)
	Polish Fisherman's Club Reef, Lorain, OH	1986–1989	Clean rock, rubble, concrete, brick scrap	Sport fishing	2 years	None	Kelch and Reutter (1995); Kelch (1997); Kelch et al. (1999); Kelch (2012)
L. Erie	The Mountain Reef, Lorain, OH	1986–1989	Clean rock, rubble, concrete, brick scrap	Sport fishing	2 years	None	Kelch and Reutter (1995); Kelch (1997); Kelch et al. (1999); Kelch (2012)
	Cuyahoga County Commissioners Reef, Lakewood, OH	1986–1989	Clean rock, rubble, concrete, brick scrap	Sport fishing	2 years	None	Kelch and Reutter (1995); Kelch (1997); Kelch et al. (1999); Kelch (2012)
	Edgewater Artificial Reef "A" (North)	1998	Old Stadium Rubble	Sport fishing	None	None	Ohio DNR (personal communication)
	Edgewater Artificial Reef "B" (South)	1998	Old Stadium Rubble	Sport fishing	None	None	Ohio DNR (personal communication)
	Euclid Artificial Reef "C"	1998	Old Stadium Rubble	Sport fishing	None	None	Ohio DNR (personal communication)
	Tawas	1987	Limestone	Fish spawning	7 years	None	Foster and Kennedy (1995)
	Thunder Bay, Cement Plant Array	2010–2011	Limestone	Fish spawning	6 years planned	–	Michigan DNR (personal communication) Healthy Lakes, 2013
	Thunder Bay CKD Pile Array	2010–2011	Limestone	Fish spawning	6 years	–	Michigan DNR (personal communication) Healthy Lakes, 2013
	Cook Nuclear Plant, Bridgman, MI		Riprap	Incidental reef	1 yr.	None	Rutecki et al. (1985)
	J.H Campbell Power Plant		Riprap	Incidental reef	2 years	None	Rutecki et al. (1985)
L. Michigan	Ludington Power Facility, MI			Incidental reef	2 years	None	Liston et al. (1985)
	South Shore Reef Park, Milwaukee, WI		Field stone/riprap	Fish spawning	2 years	3 years	Binkowski (1985); Fitzsimons (1996)
	Hamilton Reef, Muskegon, MI		Cement/rubble	Fish spawning	2 years	2 years	Kevern et al. (1985); Fitzsimons (1996)
	Port of Indiana Breakwater	1992–1996	Large and small stones (Limestone)	Incidental reef	5 years	5 years	Marsden and Chotkowski (2001)
	Chicago Waterfront	1999	Granite rubble	Sport fishing	4 years	None	Creque et al. (2006)
	Oak Creek Reef	2007	Limestone	Fish spawning	5 years	None	Yauck (2009)
	Western Lake Ontario	1981–1982	Red shale, red siltstone	Sport fishing	2 years	None	Bader (1985); Stewart and Haynes (1994)
	Stony Island	1989	Limestone rock	Fish spawning	2 years	None	Marsden et al. (1995)
	Hamilton Harbor		Rock, gravel	Fish spawning	None	None	Smokorowski et al. (1998b)
	Hamilton Harbor Shoal	2010	Rock and cobble	Fish spawning	5 years	None	Hamilton Port Authority (2010)
L. Superior	Presque Isle Harbor	1974–1977	Granite and limestone	Incidental reef	4 years	4 years	Peck (1986)
	Fitzgers Reef	1990	Boulder and Cobble	Fish spawning	3 years	None	Schreiner et al. (1995)
	Des Prairies River (St. Lawrence River tributary)	1996	Canal excavation material	Fish spawning	7 years	4 years	Dumont et al. (2011)
	Grasse River (St. Lawrence River tributary)	1996	Limestone shot rock	Fish spawning	None	None	LaPan et al. (1998)
	Current River (L. Superior tributary)	1991	Igneous rock	Fish spawning	3 years	None	Geiling et al. (1996)
	Brown County Fairgrounds (Fox River, Lake Michigan tributary)	1989	Dolomite limestone	Fish spawning	6 years	None	Lychwick (1995)
	Voyageur Park (Fox River, Lake Michigan tributary)	1990	Dolomite limestone	Fish spawning	5 years	None	Lychwick (1995)
	Fox Point Marina (Fox River, Lake Michigan tributary)	1992	Dolomite limestone	Fish spawning	3 years	None	Lychwick (1995)
Tributaries							

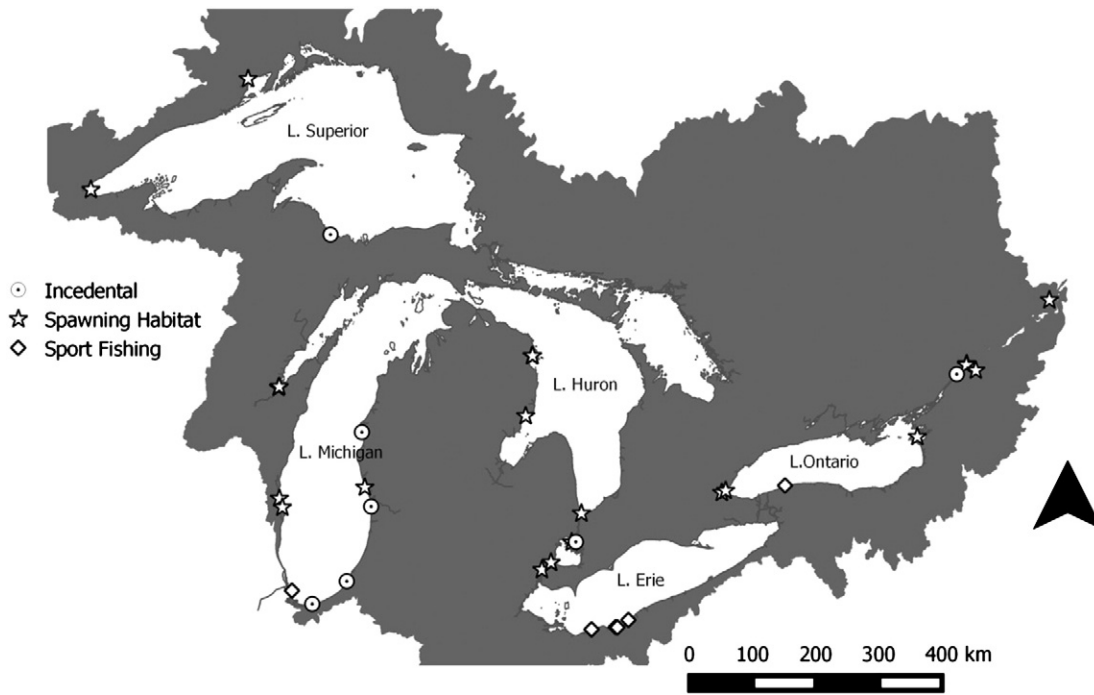


Fig. 1. Locations of the artificial reef sites discovered in the Great Lakes and Great Lakes connecting channels and tributaries. Map: Google Maps.

range of rock sizes to build artificial reefs (Geiling et al., 1996; Roseman et al., 2011). Older reefs were built in areas to provide access to anglers, but recent reefs have been built in known fish spawning areas, such as channels connecting the Great Lakes and tributary rivers.

Biological monitoring of reef projects

Biological monitoring of artificial reefs in the Great Lakes has focused on either the abundance or diversity of adult fish or on early life history stages to gauge the reproductive success of spawning fish. In general, biological monitoring of artificial reefs in the Great Lakes typically lasts for

2–7 years after reef construction (mean of 2.9 years across completed studies; Table 1). Many monitoring studies lack biological or physical assessment before reef construction to compare to post-construction conditions. Adult fish variables have included the abundance of target species, such as lake trout (Fitzsimons, 1996), total fish abundance (Kelch et al., 1999), or species richness (Kelch et al., 1999). Sampling gears and methods for adult fish have varied across reef monitoring projects and have included underwater observation (Environment Illimité Inc., 2009, 2010; Kelch et al., 1999), gill nets (Creque et al., 2006; Dumont et al., 2011), and trap nets (Geiling et al., 1996). In addition to fish sampling, surveys of fisheries biologists (Geiling et al., 1996) and

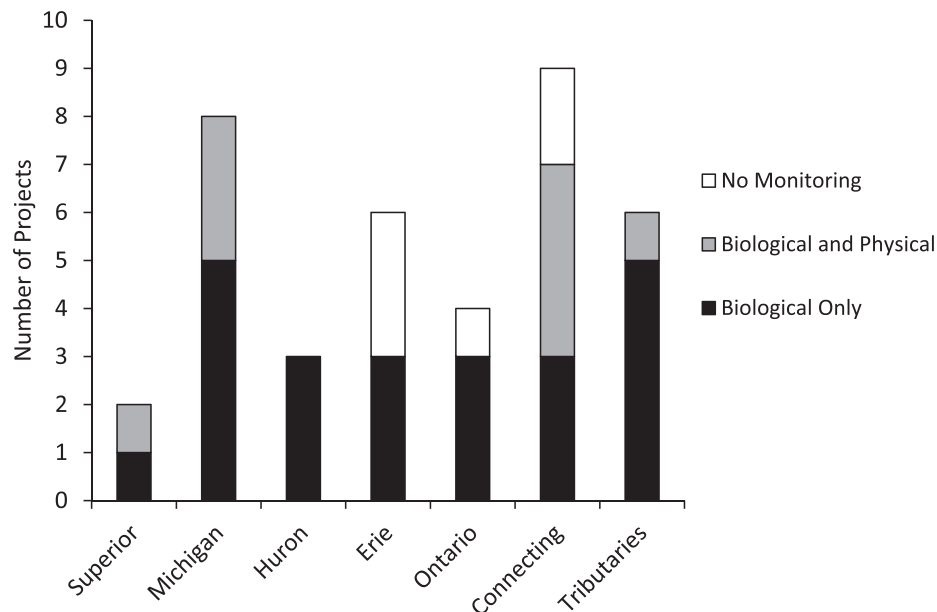


Fig. 2. Artificial reef monitoring conducted in the Great Lakes.

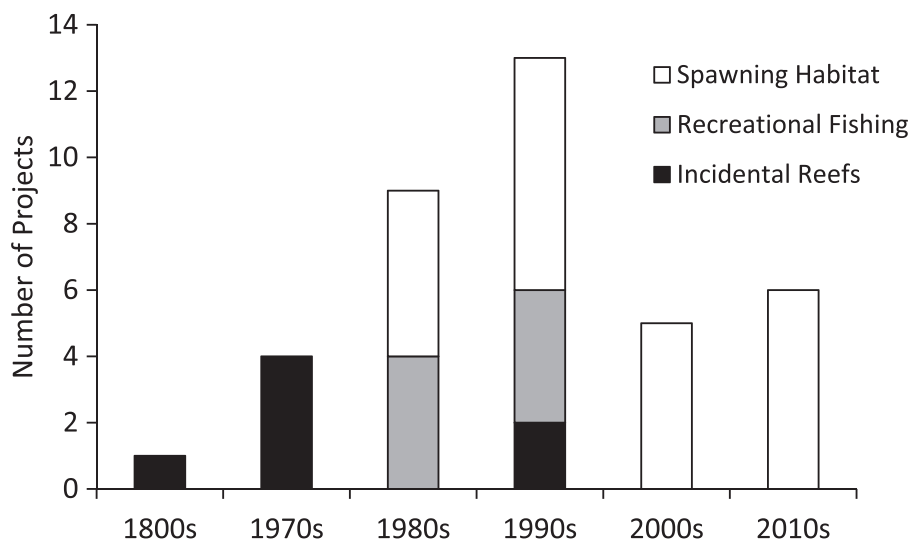


Fig. 3. Number, types, and timing of artificial reef and structure creation in the Laurentian Great Lakes.

anglers (Creque et al., 2006; Kelch, 2012) have also been used to determine adult fish use of artificial reefs.

Early life history variables have been used to measure fish production on artificial reefs and have varied among studies. For example, egg deposition has been used to assess spawning activity of lake trout on reefs in Lake Michigan and Lake Ontario (Fitzsimons, 1996) and lake sturgeon in the Detroit River (Roseman et al., 2011). Lake trout spawning activity has also been measured in Lake Michigan by the emergence of larvae from spawning substrates (Marsden and Chotkowski, 2001) and lake sturgeon spawning success has been assessed with larval fish densities in the proximity of artificial reef structures (Bouckaert et al., 2014; Roseman et al., 2011). Early life stage sampling gears and techniques have varied across studies and have included larval emergence traps (Marsden and Chotkowski, 2001), underwater observation of egg deposition (Geiling et al., 1996), egg incubation chambers (Manny et al., 1989), egg traps (Dumont et al., 2011; Roseman et al., 2011), and larval drift nets (Bouckaert et al., 2014; Roseman et al., 2011).

Physical monitoring of artificial reef projects

Most of the monitoring protocols on artificial reefs in the Great Lakes focused on monitoring the biological attributes of the reef, with the physical aspects being secondary or not measured at all (Table 1). Studies that did monitor the physical properties of artificial reefs were typically qualitative observations, made by SCUBA divers, not standardized, quantitative methods. Underwater video and remotely operated vehicles (ROV) have been used to observe zebra mussel infestation of an artificial reef in Lake Huron (Foster and Kennedy, 1995) and to evaluate sedimentation, algae, and mussel accumulation on a reef in the St. Lawrence River (Environment Illimité Inc., 2010).

Although qualitative habitat assessments have typically been used to monitor the physical aspects of Great Lakes artificial reefs, several examples of quantitative physical habitat measurements exist for artificial reefs in the Great Lakes. For example, divers have been used to map observations of a reef in Lake Michigan, including changes of reef shape over time (Binkowski, 1985). Dumont et al. (2011) quantified flow velocity, depth, and substrate on an artificial reef in the St. Lawrence River for 4 years via SCUBA. Marsden and Chotkowski (2001) described a quantitative protocol for measuring zebra mussel accumulation on an artificial reef in the Port of Indiana over a 5-year period. In a similar study, Rutecki et al. (1985) described methods for measuring algal colonization on reefs in Lake Michigan over time using SCUBA divers. Side scan sonar has also been used by researchers to monitor artificial reefs (e.g., Collier and Brown, 2004). Side scan sonar was used to evaluate

the condition of a natural reef in Lake Huron (Edsall et al., 1989, 1992), produce maps of artificial reefs in Lake Erie (Kelch, 2012) and determine the shape of an artificial reef in Lake Michigan (Kevern et al., 1985). Side scan sonar, as well as underwater video cameras, PONAR grab samples, particle size analysis, and SCUBA diver observations were all used at a proposed reef site in the Detroit River during a pre-construction assessment of a constructed fish spawning reef (Manny, 2006). However, these techniques have not been employed for post-construction assessment work. Side scan sonar was used to determine the height of reefs in Lake Erie immediately after deployment; however, no record was made of follow-up measurements (Kelch et al., 1999).

Success of artificial reefs in the Great Lakes in attracting fish and increasing fish spawning

Stated objectives of artificial reef projects constructed to enhance sport fishing include attracting higher densities of adult fish and anglers (e.g. Creque et al., 2006; Kelch et al., 1999). However, project objectives for fish spawning reefs are typically aimed at increasing fish population levels (e.g., Dumont et al., 2011; Roseman et al., 2011). These population-level objectives of spawning reefs are often measured by monitoring the number of adult spawners, egg deposition, and larval fish production. Artificial reefs have been shown to attract congregations of adult fish sought by anglers (Creque et al., 2006; Fitzsimons, 1996; Kelch et al., 1999). Moreover, fish egg deposition and survival of larvae produced have been shown to sometimes be greater on artificial reefs habitats than on natural habitats (Bouckaert et al., 2014; Dumont et al., 2011; Fitzsimons, 1996), indicating that artificial reefs could increase spawning success of Great Lakes fishes. Artificial reefs have been heavily used by numerous fish species throughout the Great Lakes and have increased fish feeding and spawning activity near reefs. Although artificial reefs have increased adult fish abundance and egg production at the site level (i.e., on reefs themselves), it is unclear whether artificial reefs have been successful in increasing year class strength of target fish populations. We found no studies that directly linked artificial reef projects to increased fish abundance at the population level (see Dumont et al., 2011; Geiling et al., 1996).

Fish have been documented to use and spawn on incidental reefs that were not intended as fish habitat. For example, scientists have documented annual spawning for years by lake sturgeon at the incidental coal cinder reef in the St. Clair River (Bouckaert et al., 2014) and lake sturgeon successfully spawned on gravel beds placed to prevent erosion of a power plant cooling water intake pipe in the St. Lawrence River (Johnson et al., 2006). Similarly, lake trout have been shown to spawn

on man-made reefs in Lake Superior (Peck, 1986), Lake Ontario (Marsden et al., 1995), and Lake Michigan (Fitzsimons, 1996). Hence, reefs built to prevent erosion, such as those routinely employed by shipping and power generation industries may also serve as fish spawning habitat, presenting an opportunity for cooperation between industry and fisheries managers (Marsden and Chotkowski, 2001).

Stressors and degradation of artificial reefs

Physical degradation of reefs can limit the effectiveness of artificial reefs as fish habitat. For example, SCUBA divers observed that silt, *Cladophora* spp., and dreissenid mussels had accumulated on a reef in the St. Lawrence River over the course of 4 years (Johnson et al., 2006). Sedimentation and algal accumulation was also observed by divers on reefs in Lake Michigan (Kevern et al., 1985), Lake Superior (Peck, 1986), and in the Detroit and St. Clair rivers (USGS unpublished data). Some reefs receive waves and currents that damage fish eggs. Others are prone to fouling by dreissenid mussels that may reduce habitat quality for fish spawning (Marsden and Chotkowski, 2001). As a result, physical stressors and degradation can limit fish egg deposition and larval fish production on artificial reefs (Johnson et al., 2006; Marsden and Chotkowski, 2001).

Limitations of artificial reef monitoring projects

Although artificial reefs have been successful in attracting fish for feeding and spawning purposes (i.e., behavioral responses), the long-term effects of artificial reefs on fishes remain unclear. Often, the intent of reef building was to increase fish populations; however, reef assessments were typically not related to broader fish community goals and objectives or to fish population and fish community changes. Most artificial reef monitoring programs are short in duration (<5 years) and small scale (<1 km²). The intended objective of many recent artificial reef projects has been to increase the population abundance of one or more fish species (e.g., Dumont et al., 2011; Fitzsimons, 1996; Roseman et al., 2011). However, lake trout, lake sturgeon, walleye, and lake whitefish are often long-lived, late-maturing, and distributed over large areas. Observance of a population-level response, such as an increase in abundance, would take much longer than the duration of most reef monitoring programs and occur over a larger spatial extent. Moreover, stressors of artificial reefs, such as sedimentation, physical degradation by waves and water currents, and changes in water levels may occur over larger spatial extents and longer time periods than most reef monitoring programs. As a result, most recent artificial reef monitoring sampling designs are inadequate to measure long-term biological responses of fish populations to constructed spawning habitat.

To measure the biological effects and permanence of artificial reefs, long-term monitoring of physical and biological attributes of constructed reefs is needed (Gannon, 1990; Hartig et al., 2009; NOAA, 2007). Long-term monitoring could potentially improve fish habitat restoration techniques and better focus limited management resources through adaptive management (Palmer et al., 2005). Inadequate monitoring of artificial reefs is common (Smokorowski et al., 1998a) and problematic to understanding the results of artificial reef projects (Baine, 2001). Long-term monitoring of man-made physical habitat was rare in the Great Lakes studies we reviewed. Furthermore, many of the reef construction projects that we reviewed lacked physical habitat monitoring or focused on a limited selection of physical habitat variables over a short time period (<5 years). The longest study that focused on the integrity of an artificial reef lasted 5 years but was narrowly focused on zebra mussel fouling (Marsden and Chotkowski, 2001). Without longer-term monitoring plans, changes in the physical and biological attributes of artificial reefs are difficult to quantify, reducing understanding of their value as fisheries management tools.

We found that reef placement, size, design, and the size and type of parent materials used to construct artificial reefs varied greatly among

the relatively small number of adequately monitored artificial reefs. Hence, our ability to discern the magnitude and significance of these confounding factors is limited. Future research is needed to determine the most effective materials and construction design needed to maximize the effectiveness of reef projects. Standardized monitoring protocols are also needed to evaluate trends in reef performance through time (Bortone and Kimmel, 1991; Murray, 1994; NOAA, 2007; Sheng, 2000). Such standardized guidelines exist for reefs constructed outside of the Great Lakes and include the focal variables and duration of monitoring (e.g., Bortone and Kimmel, 1991; Lindberg and Seaman, 2011). Although sampling methods necessarily depend on the habitat and fish species of interest, the wide variety of biological variables used to assess artificial reefs may limit transferability across reef assessment studies and limit our understanding of artificial reefs as a fisheries management practice (Fitzsimons, 1996). Historically, physical habitat monitoring of man-made reefs in the Great Lakes has typically consisted of qualitative underwater observations which may limit our ability to assess physical, temporal, and biological changes on reefs over time. More quantitative underwater observations (i.e., SCUBA divers) have been used to monitor physical attributes of man-made reefs outside the Great Lakes (e.g., Ambrose and Anderson, 1990; Wilding and Sayer, 2002). Standardized long-term monitoring of physical and biological variables on artificial reefs could improve the design and biological productivity of artificial reefs and allow fishery managers to better understand the benefits of reefs to fishery resources. We recommend that researchers focusing on Great Lakes reefs utilize quantitative methodology.

Conclusions

The use of artificial reefs as fisheries enhancement tools in the Great Lakes is widespread. Artificial reefs have been constructed in a wide range of Great Lakes habitats for many target species. Studies of artificial reefs in the Great Lakes using underwater cameras and side scan sonar advanced the monitoring of biological and physical characteristics of artificial reefs (Foster and Kennedy, 1995; Manny, 2006). However, few of these methods have been incorporated into standardized, long-term monitoring programs.

We recommend the following practices for future research on the benefits of artificial reefs to fishery resources in the Great Lakes: (1) develop reef monitoring programs that better match the spatial and temporal extent of the biological responses and physical stressors; (2) monitor the physical, as well as biological changes in reefs; (3) standardize biological and physical monitoring and methodology to allow for comparisons among reefs and through time; and (4) measure quantitative response variables, not qualitative variables of the constructed reefs. Following these recommendations should better quantify reef efficacy and performance and improve our understanding of man-made fish spawning reefs.

Acknowledgments

This report benefited from discussions with J.E. Johnson (MI DNR), A. Liskauskas (OMNR), J.H. Johnson (USGS), L. Vacarro (Michigan Sea Grant), D. Bennion, (USGS), and R. Klindt (NYDEC). P. Thompson assisted with data analysis. This work was funded by the Great Lakes Restoration Initiative Project Template 70 and the USGS Great Lakes Science Center. This is contribution number 1899 of the USGS Great Lakes Science Center.

References

- Ambrose, R.F., Anderson, T.W., 1990. Influence of an artificial reef on the surrounding infaunal community. *Mar. Biol.* 107, 41–52.
- Ambrose, R.F., Swarbrick, S.L., 1989. Comparison of fish assemblages on artificial and natural reefs off the coast of southern California. *Bull. Mar. Sci.* 44, 718–733.
- Bader, A.P., 1985. Seasonal dynamics of benthic macroinvertebrates inhabiting an artificial reef and surrounding areas in southwestern Lake Ontario. (A thesis submitted in partial fulfillment of the requirements for the Master of Science

- Degree). State University of New York, College of Environmental Science and Forestry, Syracuse, NY.
- Baine, M., 2001. Artificial reefs: a review of their design, application, management and performance. *Ocean Coast. Manag.* 44, 241–259.
- Bassett, C.E., 1994. Use and evaluation of fish habitat structures in the lakes of the eastern United States. *USDA For. Serv. Bull. Mar. Sci.* 55, 1137–1148.
- Binkowski, F.P., 1985. Utilization of artificial reefs in the inshore areas of Lake Michigan. In: D'Itri, F.M. (Ed.), *Artificial Reefs: Marine and Freshwater Applications*. Lewis Publishers, Inc., Chelsea, Michigan, pp. 349–362.
- Bombace, G., Fabi, G., Fiorentini, L., Speranza, S., 1994. Analysis of the efficacy of artificial reefs located in different areas of the Adriatic Sea. *Bull. Mar. Sci.* 55, 559–580.
- Bortone, S.A., Kimmel, J.J., 1991. Environmental assessment and monitoring of artificial habitats. In: Seaman, W., Sprague, L.M. (Eds.), *Artificial Habitats for Marine and Freshwater Fisheries*. Academic Press, San Diego, California, pp. 177–236.
- Bouckaert, E.K., Auer, N.A., Roseman, E.F., Boase, J., 2014. Verifying success of artificial reefs in the Huron-Erie Corridor for lake sturgeon. *J. Appl. Ichthyol.* 30, 1393–1401.
- Brickhill, M.J., Lee, S.Y., Connolly, R.M., 2005. Fishes associated with artificial reefs: attributing changes to attraction or production using novel approaches. *J. Fish Biol.* 67, 53–71.
- Carr, M.H., Hixon, M.A., 1997. Artificial reefs: the importance of comparisons with natural reefs. *Fisheries* 22, 28–33.
- Claudet, J., Pelletier, D., 2004. Marine protected areas and artificial reefs: A review of the interactions between management and scientific studies. *Aquat. Living Resour.* 17, 129–138.
- Collier, J.S., Brown, C.J., 2004. Correlation of sidescan backscatter with grain size distribution of surficial seabed sediments. *Mar. Geol.* 214, 431–449.
- Creque, S.M., Raffenberg, M.J., Brofka, W.A., Dettmers, J.M., 2006. If you build it, will they come? Fish and angler use at a freshwater artificial reef. *N. Am. J. Fish. Manag.* 26, 702–713.
- Dumont, P., Amours, J.D., Thibodeau, S., Dubuc, N., Verdon, R., Garceau, S., Bilodeau, P., Mailhot, Y., Fortin, R., 2011. Effects of the development of a newly created spawning ground in the Des Prairies River (Quebec, Canada) on the reproductive success of lake sturgeon (*Acipenser fulvescens*). *J. Appl. Ichthyol.* 27, 394–404.
- Edsall, T.A., Poe, T.P., Nester, R.T., Brown, C.L., 1989. Side-scan sonar mapping of lake trout spawning habitat in northern Lake Michigan. *N. Am. J. Fish. Manag.* 9, 269–279.
- Edsall, T.A., Brown, C.L., Kennedy, G.W., Poe, T.P., 1992. Lake trout spawning habitat in the Six Fathom Bank-Yankee Reef Trout Sanctuary, Lake Huron. *J. Great Lakes Res.* 18, 70–90.
- Environment Illimité Inc., 2009. Investigations of lake sturgeon spawning activities at Iroquois Dam on the St. Lawrence River in 2008—final report. Prepared for the New York Power Authority, 123 Main St. White Plains, NY.
- Environment Illimité Inc., 2010. Investigation of lake sturgeon spawning activities at Iroquois Dam on the St. Lawrence River in 2009—draft final report. Prepared for the New York Power Authority, 123 Main St. White Plains, NY.
- Fitzsimons, J.D., 1996. The significance of man-made structures for lake trout spawning in the Great Lakes: are they a viable alternative to natural reefs? *Can. J. Fish. Aquat. Sci.* 53, 142–151.
- Flopp, H., Lowry, M., Gregson, M., Suthers, I., 2013. Fish assemblages on estuarine artificial reefs: natural rocky-reef mimics or discrete assemblages? *PLoS ONE* 8, e63505.
- Foster, N.R., Kennedy, G.W., 1995. Evaluation of Tawas artificial reef as fish spawning habitat. In: Kelso, J.R.M., Hartig, J.H. (Eds.), *Methods of modifying habitat to benefit the Great Lakes ecosystem*. CISTI Occasional Paper No. 1, pp. 267–271.
- Gannon, J., 1990. International position statement and evaluation guidelines for artificial reef development in the Great Lakes. *Great Lakes Fish. Comm. Spec. Pub.* 90–92.
- Geiling, W.D., Kelso, J.R.M., Iwachewski, E., 1996. Benefits from incremental additions to walleye spawning habitat in the Current River, with reference to habitat modification as a walleye management tool in Ontario. *Can. J. Fish. Aquat. Sci.* 53, 79–87.
- Hamilton Port Authority, 2010. Lake herring and lake whitefish spawning shoal restoration in Lake Ontario. Final Programmatic Report Narrative, Sustain Our Great Lakes Stewardship Grant Report (<http://www.sustainourgreatlakes.org/Projects/GrantsAwarded.aspx>).
- Hartig, J.H., Zarull, M.A., Ciborowski, J.J.H., Gannon, J.E., Wilke, E., Norwood, G., Vincent, A.N., 2009. Long-term ecosystem monitoring and assessment of the Detroit River and Western Lake Erie. *Environ. Monitor. Assess.* 158, 87–104.
- Healthy Lakes, 2013. Artificial reefs boosting fish populations in Lake Huron. <http://healthylakes.org/uncategorized/artificial-reefs-aiding-fish-in-lake-huron/>.
- Hughes, R.M., Dunham, S., Maas-Hebner, K.G., Yeakley, J.A., Schreck, C., Harte, M., Molina, N., Shock, C.C., Kaczynski, V.W., Schaeffer, J., 2014. A review of urban water body challenges and approaches: (1) rehabilitation and remediation. *Fisheries* 39, 18–29.
- Jähnig, S.C., Lorenz, A.W., Hering, D., Antons, C., Sundermann, A., Jedicke, E., Haase, P., 2011. River restoration success: a question of perception. *Ecol. Appl.* 21, 2007–2015.
- Jelks, H.L., Walsh, S.J., Burkhead, N.M., Contreras-Balderas, S., Diaz-Pardo, E., Hendrickson, D.A., Lyons, J., Mandrak, N.E., McCormick, F., Nelson, J.S., Platania, S.P., Porter, B.A., Renaud, C.B., Schmitter-Soto, J.J., Taylor, E.B., Warren, M.L., 2008. Conservation status of imperiled North American freshwater and diadromous fishes. *Fisheries* 33, 372–407.
- Johnson, J.H., LaPan, S.R., Klindt, R.M., Schiavone, A., 2006. Lake sturgeon spawning on artificial habitat in the St. Lawrence River. *J. Appl. Ichthyol.* 22, 465–470.
- Kelch, D.O., 1997. Locating and using Lake Erie's artificial reefs. Ohio Sea Grant College Program. FS-072.
- Kelch, D.O., 2012. Lake Erie's artificial reef program: build it and they will come. Ohio Sea Grant, Fact Sheet Series FS-072.
- Kelch, D.O., Reutter, J., 1995. Artificial reefs in Lake Erie: a habitat enhancement tool. In: Kelso, J.R.M., Hartig, J.H. (Eds.), *Methods of modifying habitat to benefit the Great Lakes ecosystem*. CISTI Occasional Paper No. 1, pp. 243–249.
- Kelch, D.O., Snyder, F.L., Reutter, J.M., 1999. Artificial reefs in Lake Erie: biological impacts of habitat alteration. *Am. Fish. Soc. Symp.* 22, 335–337.
- Kevern, N.R., Biener, W.E., VanDerLaan, S.R., Cornelius, S.D., 1985. Preliminary evaluation of an artificial reef as a fishery management strategy in Lake Michigan. In: D'Itri, F.M. (Ed.), *Artificial Reefs: Marine and Freshwater Applications*. Lewis Publishers, Inc., Chelsea, Michigan, pp. 443–458.
- Koeck, B., Tessier, A., Brind'Amout, A., Pastor, J., Bijaoui, B., Dalias, N., Astruch, P., Saragoni, G., Lenfant, P., 2014. Functional differences between fish communities on artificial and natural reefs: a case study along the French Catalan coast. *Aquat. Biol.* 20, 219–234.
- Kuhl, D., 1992. 14 years of artificial grain feeding in the Rhine downstream of the barrage Iffezheim. In: Larsen, P., Eisenhauer, N. (Eds.), *Proceedings of the 5th International Symposium on River Sedimentation*. Karlsruhe, Germany, pp. 1121–1129.
- LaPan, S.R., Schiavone, A., Klindt, R.M., Colesante, R.T., 1998. 1997 Lake sturgeon restoration activities. NYSDEC Lake Ontario Annual Report 1997. St. Lawrence River Unit.
- Lindberg, W.J., Seaman, W., 2011. Guidelines and management practices for artificial reef siting, use, construction, and anchoring in Southeast Florida. Florida Department of Environmental Protection, Miami, FL (xi and 150 pages).
- Liston, C.R., Brazo, D.C., Bohr, J.R., 1985. Abundance and composition of Lake Michigan fishes near rock jetties and a breakwater, with comparisons to fishes in nearby natural habitats. In: D'Itri, F.M. (Ed.), *Artificial Reefs: Marine and Freshwater Applications*. Lewis Publishers, Inc., Chelsea, Michigan, pp. 491–514.
- Lychwick, T.L., 1995. Fox River walleye habitat improvement. In: Kelso, J.R.M., Hartig, J.H. (Eds.), *Methods of modifying habitat to benefit the Great Lakes ecosystem*. CISTI Occasional Paper No. 1, pp. 272–281.
- Manny, B.A., 2006. Monitoring element of the Belle Isle/Detroit River Sturgeon Habitat Restoration, Monitoring, and Education Project. Research completion report to Michigan Sea Grant Program. Univ. of Michigan, 401 E. Liberty, Suite 330, Ann Arbor, MI 48105.
- Manny, B.A., Jude, D.J., Eshenroder, R., 1989. Field test of a bioassay procedure for assessing habitat quality on fish spawning grounds. *Trans. Am. Fish. Soc.* 118, 175–182.
- Manny, B.A., Kennedy, G.W., Boase, J.C., Allen, J.D., Roseman, E.F., 2010. Spawning by Walleye (*Sander vitreus*) and white sucker (*Catostomus commersoni*) in the Detroit River: implications for spawning habitat enhancement. *J. Great Lakes Res.* 36, 490–496.
- Marsden, J.E., Chotkowski, M.A., 2001. Lake trout spawning on artificial reefs and the effect of zebra mussels: fatal attraction? *J. Great Lakes Res.* 27, 33–43.
- Marsden, J.E., Perkins, D.L., Krueger, C.C., 1995. Recognition of spawning areas by lake trout: deposition and survival of eggs on small, man-made rock piles. *J. Great Lakes Res.* 21, 330–336.
- Munday, P.L., 2004. Habitat loss, resource specialization, and extinction on coral reefs. *Glob. Chang. Biol.* 10, 1642–1647.
- Murray, J.D., 1994. A policy and management assessment of U.S. artificial reef programs. *Bull. Mar. Sci.* 55, 960–969.
- National Oceanic and Atmospheric Administration (NOAA), 2007. National artificial reef plan (as amended): guidelines for siting, construction, development, and assessment of artificial reefs. United States Department of Commerce (<http://www.nmfs.noaa.gov/sfa/PartnershipsCommunications/NARPWCover3.pdf>).
- Nichols, S.J., Kennedy, G., Crawford, E., Allen, J., French, J., Black, G., Blouin, M., Hickey, J., Chernyak, S., Hass, R., Thomas, M., 2003. Assessment of lake sturgeon spawning efforts in the lower St. Clair River, Michigan. *J. Great Lakes Res.* 29, 383–391.
- Palmer, M.A., Bernhardt, E.S., Allan, J.D., Lake, P.S., Alexander, G., Brooks, S., Carr, J., Clayton, S., Dahm, C.N., Follstad Shah, J., Galat, D.L., Loss, S.G., Goodwin, P., Hart, D.D., Hassett, B., Jenkinson, R., Kondolf, G.M., Lave, R., Meyer, J.L., O'Donnell, T.K., Pagano, L., Sudduth, E., 2005. Standards for ecologically successful river restoration. *J. Appl. Ecol.* 42, 208–217.
- Peck, J.W., 1986. Dynamics of reproduction by hatchery lake trout on a man-made spawning reef. *J. Great Lakes Res.* 12, 293–303.
- Perkol-Finkel, S., Shashar, N., Benayahu, Y., 2006. Can artificial reefs mimic natural reef communities? The roles of structural features and age. *Mar. Environ. Res.* 61, 121–135.
- Pickering, H., Whitmarsh, D., 1997. Artificial reefs and fisheries exploitation: a review of the 'attraction versus production' debate, the influence of design and its significance for policy. *Fish. Res.* 31, 39–59.
- Powels, H., Bradford, M.J., Bradford, R.G., Doubleday, W.G., Innes, S., Levings, C.D., 2000. Assessing and protecting endangered marine species. *ICES J. Mar. Sci.* 57, 669–676.
- Roseman, E.F., Manny, B., Boase, J., Child, M., Kennedy, G., Craig, J., Soper, K., Drouin, R., 2011. Lake Sturgeon responses to a spawning reef constructed in the Detroit River. *J. Appl. Ichthyol.* 27, 66–67.
- Rutecki, T.L., Dorr, J.A., Jude, D.J., 1985. Preliminary analysis of colonization and succession of selected algae, invertebrates, and fish on two artificial reefs in inshore southeastern Lake Michigan. In: D'Itri, F.M. (Ed.), *Artificial Reefs: Marine and Freshwater Applications*. Lewis Publishers, Inc., Chelsea, Michigan, pp. 459–489.
- Rutherford, E.S., Marshall, E., Clapp, D., Horns, W., Gorenflo, T., Trudeau, T., 2004. Lake Michigan Environmental Objectives. Available from: <http://www.glfc.org/research/reports/RutherfordEOs.pdf> [accessed 9 January 2013].
- Schreiner, D.R., Bronte, C.R., Payne, N.R., Fitzsimons, J.D., Casselman, J.M., 1995. Use of egg traps to investigate lake trout spawning in the Great Lakes. *J. Great Lakes Res.* 21, 433–444.
- Sheng, P.Y., 2000. Physical characteristics and engineering at reef sites. In: Seaman, W., Jensen, A.C. (Eds.), *Artificial reef evaluation: with application to natural marine habitats*. CRC Press LLC, Boca Raton, Florida, pp. 51–94.
- Smokorowski, K.E., Withers, K.J., Kelso, J.R.M., 1998a. Does habitat creation contribute to management goals? An evaluation of literature documenting freshwater habitat rehabilitation or enhancement projects. *Can. Tech. Rep. Fish. Aquat. Sci.* 2249.
- Smokorowski, K.E., Stoneman, M.G., Cairns, V.W., Minns, C.K., Randall, R.G., Valere, B., 1998b. Trends in the nearshore fish community of Hamilton Harbour, 1988 to 1997, as measured using an Index of Biotic Integrity. *Can. Tech. Rep. Fish. Aquat. Sci.* 2230.

- St. Clair River Bi-National Public Advisory Council, 2013. Delisting targets for loss of fish/wildlife habitat beneficial use impairment of the St. Clair River area of concern. Report to Michigan Department of Environmental Quality. Office of the Great Lakes.
- Stewart, T.W., Haynes, J.M., 1994. Benthic macroinvertebrate communities of southwestern Lake Ontario following invasion of dreissena. *J. Great Lakes Res.* 20, 479–493.
- Vincent, J., 1995. Underwater fish reef creation project within Ashbridge's Bay Park, City of Toronto. In: Kelso, J.R.M., Hartig, J.H. (Eds.), *Methods of modifying habitat to benefit the Great Lakes ecosystem*. CISTI Occasional Paper No. 1, pp. 121–127.
- Whiteway, S.L., Biron, P.M., Zimmermann, A., Venter, O., Grant, J.W., 2010. Do in-stream restoration structures enhance salmonid abundance? A meta-analysis. *Can. J. Fish. Aquat. Sci.* 67, 831–841.
- Wilding, T.A., Sayer, M.D.J., 2002. Evaluating artificial reef performance: approaches to pre- and post-deployment research. *J. Mar. Sci.* 59, S222–S230.
- Yauck, J., 2009. Oak Creek Power Plant artificial reef. Bay View Compass (Available from: <http://bayviewcompass.com/oak-creek-power-plant-artificial-reef/> [accessed 14 May 2014]).